

```
##### Testing the class #####
```

```
if __name__ == "__main__":
```

```
    # Text-book code fragment modification
```

```
    print("--- Code Fragment 2.3 Modification Test ---")
```

```
    wallet = []
```

```
    wallet.append(CreditCard('John Bowman', 'California Savings', '5391 0375 9387 5309', 2500))
```

```
    wallet.append(CreditCard('John Bowman', 'California Federal', '3485 0399 3395 1954', 3500))
```

```
    wallet.append(CreditCard('John Bowman', 'California Finance', '5391 0375 9387 5309', 5000))
```

```
    # The modified loop that pushes exactly one card over the limit:
```

```
    for val in range(1, 59):
```

```
        wallet[0].charge(val)
```

```
        wallet[1].charge(2 * val)
```

```
        wallet[2].charge(3 * val)
```

```
    for c in range(3):
```

```
        print(f"Card {c} Balance: {wallet[c].get_balance()} (Limit: {wallet[c].get_limit()})")
```

```
    print("\n--- Testing Predatory Card Features ---")
```

```
    visa = PredatoryCreditCard('Sally Shoo', 'Vells', '1234 5678 9012 3456', 5000, 0.0825)
```

```
    # Test 10+ charge surcharge
```

```
    for i in range(12):
```

```
        visa.charge(10) # Total 12 charges. Charges 11 and 12 should assess $1 surcharges.
```

```
    print(f"Balance after 12 charges of $10: {visa.get_balance()} (Expected 122)")
```

```
    # Test Late Fee
```

```
    visa.process_month() # Initial min_due was 0, so no late fee yet. New min_due generated.
```

```
    print(f"Minimum due for next cycle: {visa._minimum_due}")
```

```
    # Do not make a payment, trigger next cycle
```

```
    visa.process_month()
```

```
    print(f"Balance after ignoring payment (includes late fee & interest): {visa.get_balance()}")
```

```
    # Test ValueError on negative payment
```

```
    try:
```

```
        visa.make_payment(-50)
```

```
    except ValueError as e:
```

```
        print(f"Caught expected error: {e}")
```

```
CreditCard_Inheritance-1.py X
CreditCard_Inheritance-1.py > ...
1
2 '''Brandon Valencia Assignment 2 6/3/26'''
3 ##### Parent class of any credit card #####
4
5 class CreditCard:
6     '''A consumer credit card.'''
7
8     def __init__(self, customer, bank, acnt, limit, balance=0):
9
10         self._customer = customer
11         self._bank = bank
12         self._account = acnt
13         self._limit = limit
14
15         # Initialize the balance using the nonpublic setter method
16         self._set_balance(balance)
17
18     def _set_balance(self, b):
19         self._balance = b
20
21     def get_customer(self):
22         '''Return name of the customer.'''
23         return self._customer
24
25     def get_bank(self):
26         '''Return the bank's name.'''
27         return self._bank
28
29     def get_account(self):
30         '''Return the card identifying number (typically stored as a string).'''
31         return self._account
32
33     def get_limit(self):
34         '''Return current credit limit.'''
35         return self._limit
36
37     def get_balance(self):
38         '''Return current balance.'''
39         return self._balance
40
41     def set_limit(self, limit):
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##### Inheritance #####
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```
class PredatoryCreditCard(CreditCard):
    ''' An extension to CreditCard that compounds interest and fees '''

    def __init__(self, customer, bank, acct, limit, apr, min_payment_pct=0.05, late_fee=25, balance=0):
        ''' Create a new predatory credit card instance. '''
        super().__init__(customer, bank, acct, limit, balance)
        self._apr = apr

        # Tracking variables for monthly processing
        self._charges_this_month = 0
        self._payments_this_month = 0

        # Minimum payment metrics
        self._min_payment_pct = min_payment_pct
        self._late_fee = late_fee
        self._minimum_due = 0 # Assume 0 due for the very first month until processed

    def get_apr(self):
        ''' Returns the APR on a credit card '''
        return self._apr

    def charge(self, price):
        ''' Charge given price to the card, assuming sufficient credit limit.
            | Assesses $5 fee if denied, and $1 surcharge for >10 calls this month.
            | ...
            # Call inherited method which handles the actual incrementing and limit checks
            success = super().charge(price)

            # Assess penalty if charge is denied
            if not success:
                self._set_balance(self.get_balance() + 5)

            # Track the number of charges
            self._charges_this_month += 1

            # Assess $1 surcharge if this is past the 10th charge
            if self._charges_this_month > 10:
                self._set_balance(self.get_balance() + 1)

            return success

    def make_payment(self, payment):
        ''' Overridden to track payments made during the month. '''
        super().make_payment(payment)
        self._payments_this_month += payment

    def process_month(self):
        ''' Assess monthly interest, late fees, and reset counters. '''

        # 1. Check if the customer met the minimum payment
        if self._payments_this_month < self._minimum_due:
```